Towards a Universal Brain Model

Multi-Modal Progression Modeling

HAIM 4

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Data

Modalities: MRI, CT, Angiogram

Dataset	**Scan type**	**Target**
PhysioNet (82)	2D CT	ICH segmentation
BHSD (192 + 2200 unlabeled)	2D CT	ICH segmentation
RSNA (25k)	2D CT	ICH classification
CQ500 (491)	2D CT	ICH classification
fastMRI (7k)	MRI	–
ISLES (250)	2D CT + Tabular	Stroke Progression
APIS (100)	2D CT + MRI Pairs	Stroke
Hartford datasets (25k+)	3D CT / MRI / Angio	Temporal pairs

Objective

Learn progression of brain pathology across time and modalities

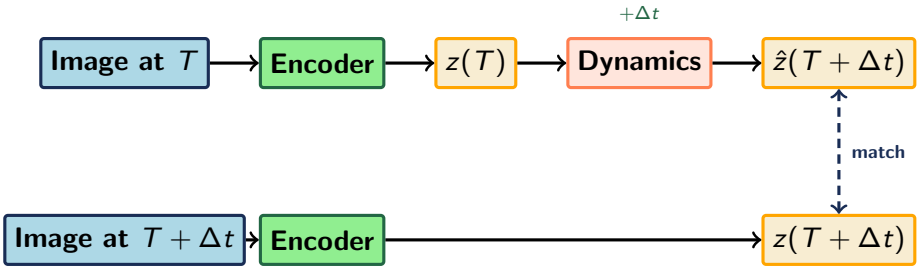
Data Challenge

- Temporal pairs from same patient are rare
- CT and MRI look completely different for same pathology

Solution: Unified latent space across modalities + dynamics model for progression

Architecture: Main

Joint-Embedding Predictive Architecture



Two key components:

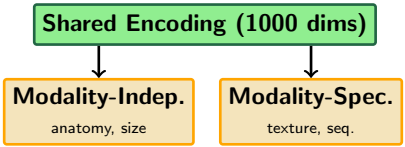
- **Encoder:** Learns shared representation across CT, MRI, angiogram
- **Dynamics:** Predicts how latent state evolves over time

Encoder & Factorized Latent Space

Self-Supervised Learning (DINO, SimCLR)

- Learns semantic features
- Robust across CT/MRI modality differences
- **Don't need labels!!!**

Factorized Latent Space: Split up information into modality indepedent and specific, only use the independent for dynamics.



Why factorize?

- Predict in independent space $\Rightarrow$ can output any modality